- There are 3 Questions for 15 marks.
- Write your answers neatly, legibly and logically. Keep your proofs thorough. Your proofs must follow from the basics discussed in class.
- You do not need any calculators.
- Good luck!

1. Consider the sequence of functions $(f_n^k : [0,1] \mapsto \mathbb{R})$

$$f_n^k(x) = \begin{cases} 2^{n/2} \text{ for } k-1 \leq 2^n x \leq k-0.5 \\ -2^{n/2} \text{ for } k-0.5 \leq 2^n x \leq k \\ 0 \text{ otherwise.} \end{cases}$$

Here $1 \leq k \leq 2^n$ and $n \geq 0$ are non-negative integers.

(a) [3] Consider an equation of the form

$$a_{n_1}^{k_1} f_{n_1}^{k_1}(x) + a_{n_2}^{k_2} f_{n_2}^{k_2}(x) + \ldots + a_{n_r}^{k_r} f_{n_r}^{k_r}(x) = 0,$$

where (n_i, k_i) are distinct pairs, and $a_{n_i}^{k_i}$ are constants/real numbers. Given the pairs (n_i, k_i), explain how to find the constants $a_{n_i}^{k_i}$.

(b) [3] How would you characterize the span of the following set of functions ?

$$\cup_{n \leq 2} \cup_{k=1}^{2^n} \{f_n^k(x)\}$$

2. Let $\mathcal{S}$ be the set of real sequences (a_n) that satisfy the following recurrence

$$a_n = a_{n-1} + a_{n-2}, n \geq 2.$$

(a) [3] Show that $\mathcal{S}$ is a vector space (with the standard addition and scaling operations) over $\mathbb{R}$.

(b) [3] What is the dimension of this vector space ?

3. [3] Is $\mathbb{C}$ a vector space over $\mathbb{R}$? If so, are 1 and i linearly independent ? What is the dimension of this vector space ?

1 (a) As discussed in class, any two functions from this collection are orthogonal.

$$\int_0^1 f_{n_1}^{k_1}(x)\, f_{n_2}^{k_2}(x)\, dx = 0 \quad \text{for any distinct } (n_1, k_1) \neq (n_2, k_2)$$

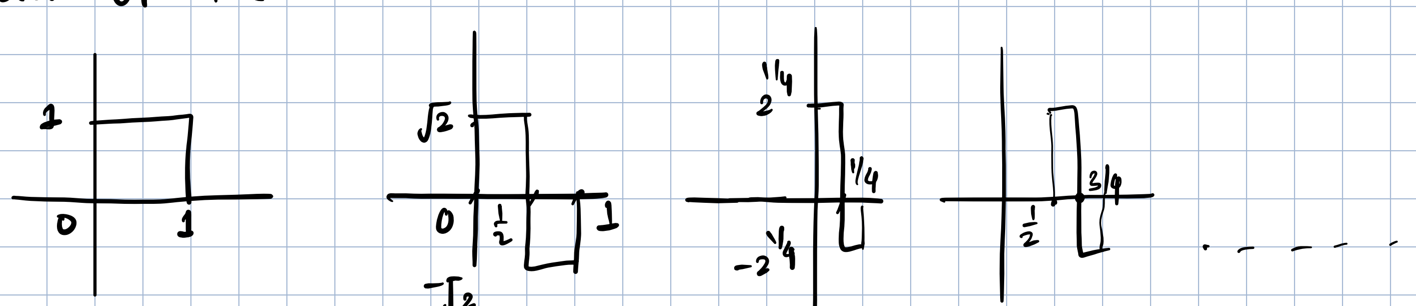
thus if $\sum_i a_{n_i}^{k_i} f_{n_i}^{k_i}(x) = 0$ — ①, then $a_{n_i}^{k_i} = 0$ for all i, since non-zero orthogonal vectors cannot sum to zero.

this can also be checked by multiplying both sides of ① with $f_{n_1}^{k_1}(x)$ and integrating both sides from 0 to 1.

$$a_{n_1}^{k_1} \int_0^1 f_{n_1}^{k_1\,2}(x)\, dx + a_{n_2}^{k_2} \cancel{\int f_{n_1}^{k_1}(x) f_{n_2}^{k_2}(x)\, dx} + a_{n_3}^{k_3} \cancel{\int f_{n_3}^{k_3}(x) f_{n_1}^{k_1}(x)\, dx} + \cdots = 0$$

(b) For $n \leq 2$, note that $f_n^k(x)$ take the same value within each of the eight intervals $(0, \frac{1}{8})$, $(\frac{1}{8}, \frac{2}{8})$, $(\frac{2}{8}, \frac{3}{8})$, ... $(\frac{7}{8}, 1)$

thus the given span is all piecewise constant functions that take same value within each of the above four intervals



$n=0$ $n=1$ $n=2$ ......

* these questions were discussed in class.

2 To check S is a vector space, we check if linear

(a) combination of two solutions is also a solution.

Let sequences (a_n) and (b_n) be solutions. Consider the linear combination sequence $c_n = \alpha a_n + \beta b_n$ for real numbers α, β.

then $c_n = \alpha a_n + \beta b_n = \alpha(a_{n-1} + a_{n-2}) + \beta(b_{n-1} + b_{n-2})$

Since a_n, b_n are solutions

$$= (\alpha a_{n-1} + \beta b_{n-1}) + (\alpha a_{n-2} + \beta b_{n-2})$$

$$= c_{n-1} + c_{n-2}.$$

So c_n is also a solution. Hence S is a vector space.

(b) Given the initial values a_0, a_1, all the a_n can be determined by the recursion. In fact, any a_n is a combination of a_0 & a_1

$$a_n = \underbrace{c_n^0}_{\text{constants}} a_0 + \underbrace{c_n^1}_{\text{(integers)}} a_1 \quad — ②$$

thus Let A_n be the solution to the recurrence when $A_0 = 0$, $A_1 = 1$ so $c_n^1 = A_n$

B_n be the solution to the recurrence when initial conditions are $B_0 = 1$, $B_1 = 0$. so $c_n^0 = B_n$

Then any solution from ② is of the form

$$a_n = B_n a_0 + A_n a_1$$

i.e every solution is linear combination of A_n & B_n

Note that A_n and B_n are independent, so the dimension is 2.

3: Yes, $\mathbb{C}$ is a vector space over $\mathbb{R}$. We can verify all the required properties (not required for grading purpose).

Consider $\alpha \cdot 1 + \beta \cdot i = 0$ for $\alpha, \beta \in \mathbb{R}$.

then $\alpha = 0$, $\beta = 0$. So $1, i$ are linearly independent

Any complex number can be written as $\alpha + \beta i$, as a linear combination of 1 & i. Since 1 & i are linearly independent, the dimension is 2.